★ Skewness :- Let X have variance σ^2 and write $m_k = E[X^k]$.
Define the skewness of X by

$$skw(X) = E[(X - m_1)^3] / \sigma^3 .$$ Show that

a) $skw(X) = (m_3 - 3m_1 m_2 + 2m_1^3)/\sigma^3$

b) $skw(S_n) = skw(X_1)/\sqrt{n}$, where $S_n = \sum_{r=1}^{n} X_r$ is a sum of independent identically distributed random variables,

c) $skw(X) = (1-2p)/\sqrt{npq}$, when X is $bin(n,p)$, where $p+q=1$,

d) $skw(X) = \frac{1}{\sqrt{\lambda}}$ when $X \overset{D.}{\sim} Poisson(\lambda)$

e) $skw(X) = \frac{2}{\sqrt{t}}$, when $X \overset{D.}{\sim} \Gamma(\lambda, t)$, and t is integral

★ Kurtosis :- Let X have variance σ^2 and $E[X^k] = m_k$.
Define the Kurtosis of X by $kur(X) = E[(X - m_1)^4] / \sigma^4$.
Show that

a) $Kur(X) = 3$, when $X \overset{D.}{\sim} N(\mu, \sigma^2)$

b) $Kur(X) = 9$, when $X \overset{D.}{\sim} exp(\lambda)$

c) $Kur(X) = 3 + \frac{1}{\lambda}$ when $X \overset{D.}{\sim} Poisson(\lambda)$

d) $Kur(S_n) = 3 + \{Kur(X_1) - 3\}/n$, where $S_n = \sum_{r=1}^{n} X_r$ is a sum of independent identically distributed random variables.